AgriPulse: Target Machine & Sensor Instrumentation Guide

Engineering Reference for Two-Stage Centrifugal Pumping System & Benchtop Prototype (v3.0)

Target Asset: Two-Stage Centrifugal Water Pump	**Telemetry Archive:** IBM/Kaggle (220,320 rows)
Standard Ref: ISO 10816-3, ANSI/HI 9.6, ISA-18.2	**Deployment Target:** Railway Cloud & ESP32 IoT Gateway

1. Target Asset Architecture & Mechanical Provenance

The AgriPulse predictive maintenance system is trained on high-resolution continuous telemetry from an industrial **Two-Stage Centrifugal Pumping System**. Telemetry encompasses 220,320 consecutive one-minute records (April–August 2018) containing 7 recorded failure events. The asset features a three-phase motor driving a two-stage pump casing via a flexible mechanical coupling.

2. Core-7 Sensor Instrumentation & Physics Mapping

Through empirical feature ablation on historical failure events, a Core-7 sensor group was established, monitoring the primary mechanical, electrical, and hydraulic stress nodes:

Sensor ID	Monitored Location	Physical Quantity & Failure Mode	Nominal Baseline
sensor_00	Motor DE Bearing	Radial vibration, race fatigue, rotor unbalance	~1.4 - 2.5
sensor_04	Motor Drive Shaft	Rotational speed proxy, motor slip (CPM)	~630 - 640
sensor_05	Motor Stator Feed (A)	Current draw, winding electrical load	~70 - 80
sensor_11	Motor Stator Feed (B)	Secondary current tap, phase balance	~45 - 55
sensor_13	Flexible Shaft Coupling	Axial vibration, angular/parallel misalignment	~1.5 - 3.2
sensor_28	Stage-2 Impeller	Stage 2 rotational speed / cavitation proxy	~1020 - 1080
sensor_40	Thrust Bearing Housing	Thrust bearing temperature, lubrication failure	~50 - 65 °C

3. Directional Multi-Sensor Diagnostic Signatures

Root cause isolation uses directional multi-sensor pattern matching rather than isolated single-sensor thresholds:

Fault Category	Sensor Signature	Physical Mechanism & Action
Coupling Misalignment	sensor_13 (↑) + sensor_00 (↑)	Simultaneous axial & DE radial vibration surge indicates coupling misalignment or degraded elastomeric insert.
Hydraulic Binding	sensor_05 (↑) + sensor_28 (↓)	Current draw surges while impeller speed decelerates, confirming internal impeller obstruction or closed valve.
Lubrication Starvation	sensor_40 (↑) with normal vib	Isolated thermal rise without mechanical vibration indicates grease dry-out or degradation.
Bearing Raceway Fatigue	sensor_00 (↑) isolated	DE radial vibration elevation with stable coupling indicates localized bearing spalling/flaking.

Decoupling / Flatline	std_60 < 1e-3 (variance collapse)	Addresses Event 5 blindspot: flags sensor disconnection or zero-variance failure immediately.

4. Industrial Alarm Debouncing (ANSI/ISA-18.2)

To suppress transient electrical spikes and reduce false alarms, the real-time engine enforces stateful time persistence:

Alarm Tier	Calibrated Cutoff	Debounce Requirement	Verified Operational Lead Time
WARNING	0.3017 (June 95.0%)	5 Consecutive Minutes	Up to 120 minutes prior to failure
CRITICAL	0.3691 (June 98.5%)	8 Consecutive Minutes	46 min (Event 6), 70 min (Event 7)

5. Benchtop Scale Prototype (12V DC Pump Rig)

A physical test bench demonstrates the real-time IoT-to-cloud acquisition chain using an ESP32 microcontroller and a 12V DC circulating pump loop. Telemetry is adapted dynamically using relative Z-score normalization:

Channel / Role	Component	Interface / Address	Bench Operating Range
Motor Vibration	ADXL345 Accelerometer	I2C (0x53)	0.1 to 0.8 g RMS
Coupling Vibration	ADXL345 Accelerometer	I2C (0x1D, SDO to 3.3V)	0.05 to 0.5 g RMS
Motor Current	ACS712 30A DC Module	ADS1115 A0 (Analog)	1.8 to 3.5 A DC
Surface Temp	DS18B20 Waterproof Probe	1-Wire (GPIO 4)	25 to 45 °C
Discharge Pressure	G1/4" Transducer (1.2 MPa)	ADS1115 A1 (Analog)	0.4 to 0.9 bar
Water Flow Rate	YF-S201 Turbine Meter	Pulse Interrupt (GPIO 34)	3.0 to 7.5 L/min

Engineering Notice: *The prototype demonstrates methodology transfer (streaming feature calculation, temporal debouncing, and multi-sensor fault isolation) rather than direct weight transfer. When running in prototype mode (mode: 'prototype'), the engine automatically employs self-calibrating relative deviations.*